Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/12/2005
PI Name:	Mulloth, Lila M M.S.		
Project Title:	Development of a Solid-State Compressor for Closed-Loop Air Revitalization Systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	650-604-1904
Organization Name:	NASA Ames Research Center		
PI Address 1:	M/S 239-15		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	02/01/2003	**End Date:**	02/01/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	John E. Finn		
COI Name (Institution):	LeVan, Martin Douglas Ph.D. (Department of Chemical Engineering, Vanderbilt University) Knox, James C (NASA Marshall Space Flight Center)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The temperature-swing adsorption compressor (TSAC) is a solid-state compressor that has the capability to remove CO2 from a low-pressure source and subsequently store, compress, and deliver at a higher pressure as required by a processor. The TSAC is an ideal interface device for CO2 removal and reduction units in the air revitalization loop of a spacecraft such for oxygen recovery. The concept of TSAC as a potential interface for the carbon dioxide removal and reduction units in a life support system has already been proven and its operation has been demonstrated The ultimate objective of the task is to build an Engineering Development Unit (EDU) of an air-cooled TSAC with flight-like components and configuration and to test it in a relevant environment for certifying the technology readiness level of 6 (TRL 6) for this device. This two-bed prototype of TSAC will have a configuration that permits continuous acquisition of CO2 from a low-pressure source such as CDRA and continuous supply of CO2 to the Sabatier reactor at the specified		

	pressure, and will be sized to support a 4-person crew load in the ISS. The project involves three phases such as thermal model development and validation, full prototype design and testing, and EDU development This technology is important to closure of the oxygen loop on International Space Station which may offer substantial cost savings due to reduced logistics. Impact on future missions is likely to be higher
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	There are three phases for this project with the following targets. Phase I: Development of the thermal model, mechanical design, and model validation (testing) of the air-cooled TSAC Phase II: Fabrication and testing of a full-size prototype to process 4kg CO2 per day Phase III: Develop an engineering development unit (EDU) with a technology readiness level (TRL) 6. Phase I targets were met on schedule on December 15, 2003. Test results indicated that efficiency of the compressor exceeded the design specification by about 50% with less than 30% of the estimated power. Results of the thermal model validation experiments were presented at the Habitation 2004 Conference at Orlando, Florida on January 5 and at the ALS workshop at Gainesville, Florida on January 9. During Phase II, fabrication of the full-size prototype and initial testing were completed on schedule, on June 30. Results of full-size system design and initial test data were presented at the International conference on Environmental Systems, Colorado Springs, Colorado on July 20. The compression tests were delayed because of leaks in the compressor chamber due to a faulty fabrication process implemented by the vendor (during the dip brazing process). The faulty parts are being re-manufactured and the compression tests were completed successfully on December 31, 2004. Development of the EDU has been initiated.
Bibliography Type:	Description: (Last Updated: 04/24/2007)
Papers from Meeting Proceedings	Mulloth LM, Affleck DL, Rosen M. "Air-Cooled Design of a Temperature-Swing Adsorption Compressor for Closed-Loop Air Revitalization Systems." 34th International Conference on Environmental Systems, Colorado Springs, Colorado, July 2004. SAE Paper No. 2004-01-2374, July 2004. , Jul-2004
Papers from Meeting Proceedings	Knox JC, Mulloth LM, Affleck DL. "Integrated Testing of a 4-Bed Molecular Sieve and Temperature-swing Adsorption Compressor for Closed-Loop Air Revitalization." 34th International Conference on Environmental Systems, Colorado Springs, Colorado, July 2004. 34th International Conference on Environmental Systems, July 2004. , Jul-2004
Papers from Meeting Proceedings	Mulloth LM, Rosen M, Affleck L, LeVan MD, Wang Y. "Development and Testing of a Temperature-Swing Adsorption Compressor for Carbon Dioxide in Closed-Loop Air Revitalization Systems." 35th International Conference on Environmental Systems (ICES), Rome, Italy, July 2005. Paper No. 2005-01-2941, 05ICES-199, July 2005. , Jul-2005
Papers from Meeting Proceedings	Rosen M, Mulloth LM, Varghese M. "Long-Duration Testing of a Temperature-Swing Adsorption Compressor for Carbon Dioxide for Closed-Loop Air Revitalization Systems." 35th International Conference on Environmental Systems (ICES), Rome, Italy, July 2005. 05ICES-203, July 2005. , Jul-2005
Presentation	Wang, Y., LeVan, M. D., Cavalcante, C. L., Mulloth L. M., Affleck, D. L., Rosen M. " Temperature-swing Adsorption Compressor (TSAC) for Closed-loop Air Revitalization System" Fundamentals of Adsorption (FOA8), Sedona, Arizona, May 23-28, 2004 (poster) May-2004